ORDI: Orbit-Aware Redundant Distributed Inference for LEO Earth Observation Constellations

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Abstract—Low-Earth-orbit Earth-observation constellations collect more imagery than intermittent ground contacts can promptly downlink, motivating distributed onboard inference for time-critical alerts. Yet changing contact geometry, thermal and battery throttling, and correlated orbital-plane failures make static task placement unreliable. We present ORDI, an orbit-aware rolling-horizon scheduler that jointly considers predicted contacts and each satellite's compute, energy, temperature, queue, and availability state. ORDI ranks image tiles by urgency-weighted utility, selects feasible primary helper–aggregator pairs, and adds fault-disjoint backups only when their marginal reliability gain exceeds energy, communication, and replication costs. Detected failures, missed contacts, and stragglers trigger local replanning. We compare ORDI with eight baselines across fault type and intensity, deadline, constellation scale, correlated outage, and backup-cap studies, using a uniform conservative 25° ground-link elevation mask. Under nominal stochastic loss, ORDI achieves the highest utility while using 40% fewer replicas and 40% less energy than full replication. Under one-plane outages it remains within 3.5 percentage points of full replication's realized miss ratio with 44% less energy. Its greedy solution attains slightly higher utility than an ILP reference with 5.8× less inter-satellite traffic. Tests on real-world NASA FIRMS wildfire requests further preserve the synthetic study's utility ordering, supporting selective redundancy under operational demand conditions.

Keywords—LEO constellations, Earth observation, distributed inference, fault-tolerant scheduling

I. Introduction

Earth observation constellations now produce imagery at rates that overwhelm their ground segments. A single satellite can capture tens of gigabytes per orbit, yet sees a ground station for only a few minutes per pass, and smaller operators often lease a handful of stations concentrated at high latitudes. The result is a widening gap between what is sensed and what can be delivered before it loses value: a wildfire alert is useful within minutes, not hours [1], [2].

Onboard inference attacks this gap at the source. Instead of downlinking raw images, satellites run compact neural networks (e.g., MobileNetV2, YOLOv5-nano) and downlink only labels, bounding boxes, or masks, a reduction of three to four orders of magnitude in volume [3]. Recent in-orbit experiments on the Tiansuan constellation demonstrate that COTS accelerators can run such models in orbit, but also document the operational reality: computing capacity is throttled by thermal limits and battery state, and the energy available per orbit is scarce and variable [4].

A natural next step is to distribute inference: a source satellite partitions an image into tiles and offloads tiles over inter-satellite links (ISLs) to less-loaded neighbors, meeting deadlines that onboard compute alone cannot. Prior systems explore single-satellite query bifurcation [2] and multi-satellite placement [5], but largely assume that scheduled helpers and links remain available. In orbit this assumption routinely fails, and it fails in structured ways: an ISL outage severs all routes through a plane, a plane-wide bus anomaly takes down every satellite sharing it, and thermal throttling turns a scheduled helper into a straggler mid-task. Classic terrestrial answers, replicate everything [6] or apply MDS-coded redundancy [7], [8], translate poorly: every replica costs scarce ISL bandwidth and battery energy, and coding assumes independent failures that orbital geometry violates.

This paper asks: how much redundancy does distributed orbital inference actually need, and where should it be placed? Our answer is ORDI (Orbit-aware Redundant Distributed Inference), a rolling-horizon scheduler built on three ideas.

(1) Orbit-aware feasibility. ORDI plans on a time-expanded contact graph derived from orbital propagation, jointly with per-satellite state: compute rate, battery energy, temperature, queue, and availability. A candidate assignment of a tile to helper i with aggregator a is feasible only if its end-to-end latency, transfer to the helper, compute, relay to the aggregator, and downlink, fits the remaining deadline budget under current link and node state.

(2) Selective, fault-disjoint redundancy. Each tile receives one primary replica chosen by marginal utility, and at most one backup, placed only when the marginal gain in delivery probability, discounted by a freshness decay and priced against energy, bandwidth, and an explicit replication penalty λ_R, is positive. Backups must use a different helper and a different aggregator than the primary, and under correlated-failure threat models, a different orbital plane, so that a single structural failure cannot eliminate both replicas.

(3) Reactive replanning. Detected helper failures, missed contacts, and stragglers trigger re-scheduling of affected tiles within the epoch, using the same feasibility machinery.

We implement ORDI and eight baselines in a discrete-epoch simulator with a realistic LEO-EO configuration: a

Walker constellation at 550 km, field-of-view-constrained task arrivals, per-workload log-normal deadlines reflecting EO service tiers [9], and satellite power, thermal, and compute parameters loaded directly from the public in-orbit measurement logs of a Tiansuan COTS payload [4]. Experiments over up to 60 random seeds show that ORDI (i) outperforms non-distributed baselines on deadline miss ratio and delivered utility, (ii) retains 77% more delivered utility than non-redundant placement at 50% fault intensity, (iii) stays within 3.5 percentage points of full replication's realized miss ratio under a one-plane outage with 44% less energy, and (iv) slightly exceeds the ILP reference's utility while using 5.8× less ISL traffic.

II. Background and Related Work

A. Onboard Inference and Orbital Edge Computing

EO satellites are bandwidth-starved: ground contact windows last only 5–10 minutes per pass, and many operators schedule only high-latitude stations, so a satellite over the southern hemisphere may wait most of an orbit to downlink. Onboard inference attacks this at the source, reducing a megabyte-scale tile to a kilobyte-scale result and shifting the bottleneck from downlink to onboard compute, which is in turn bounded by a small spacecraft's power and thermal envelope. That this is practical is now established by flight programs across agencies and industry: ESA's Φ-Sat-1 flew the first on-board deep-neural-network demonstrator [10], re-trainable payloads have been uplinked and run in orbit on a 1 W VPU [11], the NASA/HPE Spaceborne Computer-2 ran AI workloads on the ISS with software radiation hardening [12], and the Tiansuan constellation reported systematic in-orbit COTS measurements [4] showing accelerators throttled well below nameplate throughput by thermal and battery limits. Earlier work established the paradigm (orbital edge computing [1], onboard model specialization [3]), and recent surveys catalogue it [13]. This body of work is almost entirely single-satellite inference; ORDI addresses the next question of distributing one task across satellites when a single satellite's compute or downlink cannot meet the deadline.

B. Multi-Satellite Offloading and Placement

As LEO mega-constellations mature, several systems schedule computation across satellites. Serval bifurcates near-real-time EO queries on one satellite's resources [2]; SECO optimizes latency or energy for multi-satellite placement [5]; and LEOEdge co-designs satellite-ground AI-inference placement with per-satellite adaptive models [14]. These approaches optimize placement under nominal availability but assume scheduled resources survive; we show this assumption costs 3.3× in miss ratio at realistic fault rates, and that structured redundancy closes the gap cheaply.

C. Priced Redundancy under Correlated Failures

Tiling makes inference embarrassingly parallel: a 4×4 grid yields 16 independent units that ISLs can spread across a neighborhood. But every replica's success requires the source to emit the tile, each ISL hop to hold, the helper to survive compute, and the aggregator's downlink to materialize—and these events are correlated, since satellites in one plane share geometry, eclipse cycles, and deployment batch, so a plane-wide outage disables many helpers at once. Tail-tolerant duplication [6] and coded computation [7], [8] address this in datacenters, but assume cheap bandwidth and independent failures. In orbit, full replication costs roughly double the energy and ISL traffic that a 20–30 Wh-class spacecraft cannot afford, and correlation breaks the coding assumption. The design space between zero redundancy and full replication, priced by orbital state and placed against the actual failure correlation, is the territory ORDI explores.

III. System Model

A. Network and Contact Model

We consider N satellites $\mathcal{S}$ and ground stations $\mathcal{G}$ over 60 s epochs. We set h=550 km, within PlanetScope's 450–580 km Earth-observation orbit range [15]. An optical ISL exists at slant range $d \leq 4{,}000$ km, the limit specified for Starlink mini lasers [16]. At this altitude, the threshold is 26% below the 5,408 km geometric line-of-sight limit and gives 255 km minimum limb clearance. All benchmarks use a conservative 25° RF elevation mask, matching an authorized minimum for LEO user and gateway links at comparable 540–570 km altitudes [17]. Contacts form per-epoch graphs $E(t)$, and earliest-arrival Dijkstra search over the time-expanded graph yields $\ell_{ab}(t, d)$, including store-and-forward waits.

B. Satellite State

Each satellite i carries state $\sigma_i(t) = (C_i, B_i, \Theta_i, Q_i, A_i)$: compute rate (GFLOP/s), battery energy (J), temperature, queued work, and availability. Battery follows an energy balance of solar harvest against idle, compute, and communication draw; temperature follows a first-order RC model driven by compute-power dissipation. Availability $A_i = 0$ when battery falls below a floor, temperature exceeds a ceiling, or a failure is injected; thermal stress throttles C_i before forcing unavailability. Rather than assuming datasheet values, we instantiate σ_i parameters (compute rate, idle and active power, battery capacity, solar harvest, thermal limits) from public in-orbit measurement logs of an Atlas 200DK payload on the Tiansuan BUPT-1 satellite [4].

C. Task and Tile Model

Tasks arrive as a Poisson process, gated by sensing geometry: a task is generated only when a satellite's camera footprint covers a ground target. Task k released at r_k on source satellite s_k carries an image partitioned into tiles v with input size d_{kv}^{in}, output size d_{kv}^{out}, compute demand c_{kv} (FLOPs), and utility u_{kv}. We model four EO workloads profiled on MobileNetV2, YOLOv5-nano, a lightweight U-Net, and a Siamese CNN (wildfire detection, ship detection, cloud filtering, change detection), with per-type deadlines drawn log-normally (σ=0.6) around medians of 300, 600, 1200, and 480 s respectively, reflecting EO service tiers [9].

D. Reliability Model

Links and nodes succeed per epoch with probabilities π: ISL π_{isl}=0.97, downlink π_{down}=0.92 (0.70 adverse), node survival π_{node}=0.98. A replica of tile (k, v) on helper i with aggregator a succeeds with

$$p_{kvia} = \pi_{s_k}\, \pi_i \cdot \pi_{\text{path}}(s_k \to i) \cdot \pi_{\text{path}}(i \to a) \cdot \pi_{\text{down},a}. \quad (1)$$

Source survival is a single point of failure shared by all replicas of a tile, so it is factored once at the tile level: with replica set R_{kv},

$$z_{kv} = \pi_{s_k} \Big[1 - \prod_{r \in R_{kv}} \big(1 - p_r / \pi_{s_k} \big) \Big], \quad (2)$$

which avoids overcounting redundancy against a failure no replica can survive. Independence across the remaining terms is an explicit approximation; Section V-I stresses it with correlated plane-wide faults.

E. Problem Statement

Let L_{kv} denote the earliest completion latency among a tile's replicas and $\tau_k(t)$ the remaining deadline budget. The scheduler chooses replica sets to maximize

$$\sum_{k,v} u_{kv}\, z_{kv}\, e^{-\alpha L_{kv}} \;-\; \lambda_E E \;-\; \lambda_C C \;-\; \lambda_R R, \quad (3)$$

subject to per-replica deadline feasibility $L_{kvia} \le \tau_k(t)$, helper energy budgets $B_i \ge B_{\min}$, and link capacities, where E is total helper energy, C is congestion-weighted ISL traffic, R is the count of extra replicas, and $e^{-\alpha L}$ encodes freshness decay (α=0.002 s^{-1}). The exact problem is an integer program coupling routing, placement, and redundancy; we use it (Section V-J) only as a small-instance reference.

IV. ORDI Design

Our code is available at https://github.com/namwoam/ordi. ORDI is a rolling-horizon greedy scheduler (Algorithm 1). At each epoch it reads the current compute, battery, thermal, queue, and availability state, builds earliest-arrival routing caches from the time-expanded contact graph, and orders pending tiles by urgency-weighted utility u_{kv}/τ_k. This ordering gives scarce compute and contact capacity to valuable tiles with little remaining slack.

For tile (k, v), an available helper–aggregator pair (i, a) is a candidate with latency

$$L_{kvia} = \ell_{s_k i}(d_{kv}^{\text{in}}) + \frac{c_{kv}}{C_i(t)} + \ell_{ia}(d_{kv}^{\text{out}}) + \ell_a^{\text{down}}(d_{kv}^{\text{out}}), \quad (4)$$

and is feasible iff $L_{kvia} \le \tau_k(t)$, $A_i = A_a = 1$, and its committed energy leaves the helper above $B_{\min}$. The set also includes helper-as-aggregator and source self-processing cases. The four latency terms reduce to source-to-all, all-pairs, and all-to-ground earliest-arrival lookups; shared Dijkstra sweeps compute these dense caches once per epoch rather than once per candidate.

Each feasible candidate is scored by its primary-placement gain

$$\Delta_{kvia} = u_{kv}\, p_{kvia}\, e^{-\alpha L_{kvia}} - \lambda_E \Delta E_{kvia} - \lambda_C \Delta C_{kvia}, \quad (5)$$

where ΔE includes receive, compute, and transmit energy and ΔC prices ISL bits by inverse link capacity. The highest-scoring candidate becomes the primary. The remaining candidates are scanned in score order; a candidate with a different helper and aggregator (and, under a correlated-failure model, a different orbital plane) is admitted as the backup only if

$$u_{kv}\, \Delta z_{kv}\, e^{-\alpha L} - \lambda_E \Delta E - \lambda_R > 0, \quad (6)$$

where Δz_{kv} is the reliability increase from Eq. (2). Thus reliable primaries remain unreplicated, while uncertain paths receive one fault-disjoint backup when its freshness-discounted benefit covers its cost. Committing a replica updates the epoch's energy and link budgets before the next tile is considered.

In Algorithm 1, H_R is the set of selected helpers and $\phi(h)$ returns helper h's orbital plane. PlaneDisjoint is enabled only for the correlated plane-outage model in Section V-I. $\Delta(r \mid R)$ is the marginal gain in Eq. (6). The default $b_{\max}$=1 implements selective single-backup redundancy; Section V-H ablates larger caps. Routing costs $O(n{\cdot}\text{SSSP})$ per epoch and enumeration costs $O(mn^2)$ cached lookups, whereas the ILP is limited to small instances.

V. Evaluation

A. Setup

Unless stated otherwise: Walker constellation, 6 planes × 6 satellites at 550 km (96 min period); horizon of 3 orbits (17,280 s, 288 epochs); two northern ground stations (Fairbanks, Greenwich), a deliberately constrained ground segment typical of high-latitude EO operations that creates genuine routing pressure for southern-hemisphere sources; field-of-view-gated Poisson arrivals at 6 tasks/orbit over 100 ground targets; 25° ground-station elevation mask; ISL rate 200 Mb/s. Satellite parameters are the Tiansuan-calibrated COTS profile. The core comparison further stresses this setup with a sparser 6×4 Walker (24 satellites) and 8 tasks/orbit, so that deadlines bind and routing pressure is real; it aggregates 60 random seeds. Fault and sweep experiments use matched seeds across algorithms; error bars are one standard deviation. Table VI in the appendix gives every per-experiment configuration.

Algorithm 1 ORDI rolling-horizon scheduler

Require: epoch t, pending tiles $\mathcal{P}$, states $\sigma(t)$, contacts $E(t)$, backup cap $b_{\max}$, PlaneDisjoint

Build earliest-arrival cache $\Lambda(t)$ from $E(t)$
Sort $\mathcal{P}$ by decreasing $u_{kv}/\tau_k(t)$
for $(k, v) \in \mathcal{P}$ **do**
  $\mathcal{F}_{kv} \leftarrow$ feasible replicas satisfying Eq. (4), availability, and energy constraints
  if $\mathcal{F}_{kv} = \emptyset$ **then**
    continue
  end if
  $r_1 \leftarrow \arg\max_{r \in \mathcal{F}_{kv}} \Delta_r$; $R \leftarrow \{r_1\}$
  $\mathcal{D} \leftarrow \mathcal{F}_{kv} \setminus R$; sort $\mathcal{D}$ by decreasing Δ_r
  while $\mathcal{D} \neq \emptyset$ and $|R| - 1 < b_{\max}$ **do**
    Remove candidates sharing a helper or aggregator with R
    if PlaneDisjoint **then**
      $\mathcal{D} \leftarrow \{r \in \mathcal{D} : \phi(h_r) \notin \phi(H_R)\}$
    end if
    $r_b \leftarrow$ first $r \in \mathcal{D}$ with $\Delta(r \mid R) > 0$
    if r_b does not exist **then**
      break
    end if
    $R \leftarrow R \cup \{r_b\}$; $\mathcal{D} \leftarrow \mathcal{D} \setminus \{r_b\}$
  end while
  Commit R; update energy and link usage
end for
if failure, missed contact, or 1.5× straggler **then**
  Mark failed helpers unavailable
  Reschedule only affected tiles within epoch t
end if

Baselines. B1 direct downlink (bent pipe, no inference); B2 onboard-only; B3 onboard with 0.15× output compression; B4 Serval-like priority bifurcation, one satellite per task [2]; B5 SECO-like multi-satellite placement without redundancy [5]; B6 full replication to $r_{\max}$ helpers [6]; B7 random-helper replication; B8 CoCoI-like MDS-coded redundancy adapted to contact windows [7], [8]. All baselines share ORDI's contact graphs, state model, and routing caches.

Metrics. Deadline miss ratio (primary), delivered utility per (3)'s first term, partial task coverage, recovery latency, ISL traffic, energy, and mean replicas per tile. Energy includes onboard compute, ISL communication, and spacecraft-to-ground transmission; the latter is charged as transmit power times airtime at the modeled 100 Mbps downlink rate, consistent with demonstrated high-rate CubeSat links surveyed by NASA [18]. Thus B1 pays to transmit raw tile inputs, while inference methods transmit compact outputs.

B. Core Comparison

Table I and Fig. 1 summarize. Direct downlink misses 95.1% of deadlines, confirming that bent-pipe delivery is not viable with this constrained ground segment. Onboard-only methods (B2–B4) miss 85.3%, limited by source compute and source-local downlink geometry. Distributed schedulers reduce misses to about 70.5%. Within that group, ORDI achieves the highest utility (26.0±14.7) at 0.32 replicas/tile and 103 J. Full replication uses 0.54 replicas and 172 J—about 67% more energy and 2.5× the ISL traffic—yet delivers less utility (24.4). The sub-unit replica averages include infeasible tiles for which no placement can meet the deadline.

TABLE I
Core comparison results, mean over 60 seeds.

Algorithm	Miss ratio	Utility	Replicas	Energy (J)
B1 direct	0.951	8.0	0	0.41
B2 onboard	0.853	17.2	0.15	45.6
B3 compress	0.853	17.6	0.15	4.7
B4 Serval-like	0.853	17.2	0.15	45.6
B5 SECO-like	0.705	22.8	0.30	94.8
B6 full repl.	0.705	24.4	0.54	171.9
B7 random repl.	0.705	23.6	0.55	175.4
B8 CoCoI-like	0.705	24.5	0.55	174.9
ORDI	0.702	26.0	0.32	103.0

C. Robustness Across Fault Types

Injecting each of eight fault classes against ORDI (Fig. 2), ground-contact misses, adverse downlink, battery shortage, thermal throttling, and stragglers leave the planned miss ratio at the no-fault value of 53.3%. ISL disruption raises it to 59.1%; plane outage and helper failure are most damaging at 61.7%. Stragglers do not change feasibility but increase energy from 119 to 218 J as replanned duplicates complete alongside slow originals.

D. Graceful Degradation Under Fault Intensity

Sweeping random-fault intensity from 0 to 50% (Fig. 3), B5's miss ratio rises from 51.2% to 81.2%, while ORDI reaches 77.5%. The miss-ratio gap is modest under the restrictive contact mask, but ORDI retains substantially more value: at 50% faults it delivers 28.5 utility versus B5's 16.1. Its replica count falls from 0.56 to 0.31 because heavy faulting removes feasible backups. Full replication holds misses near 53% by consuming roughly three times ORDI's energy at 50% intensity.

E. Scalability

From 12 to 60 satellites (Fig. 4), misses fall from 89.8% to 4.0% as contact and helper density improve. ORDI and B5 see the same planned feasibility, so their miss ratios coincide; ORDI delivers 7–16% more utility across the sweep through latency-aware placement and selective backups.

F. Deadline Sensitivity

Sweeping the global deadline scale from 150 to 900 s (Fig. 5), ORDI beats the single-satellite schedulers at

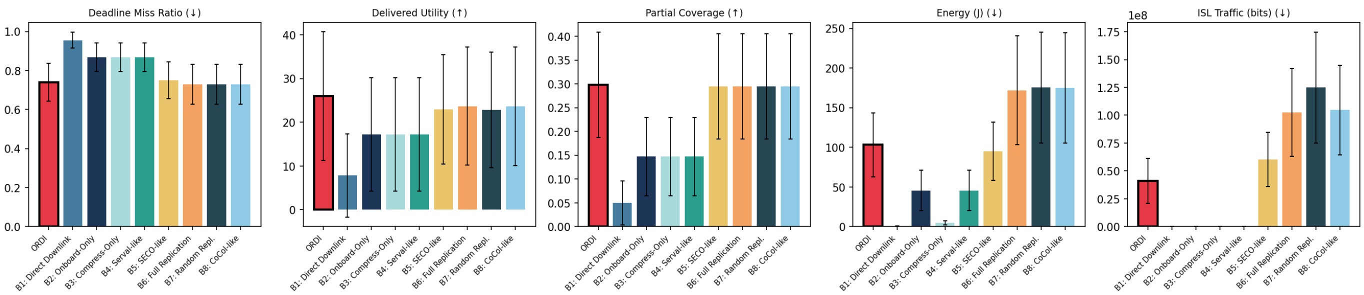

Fig. 1. Core comparison over 60 seeds (no injected faults beyond baseline stochastic link/node loss). ORDI attains the best delivered utility and ties the best miss ratio at near-minimal redundancy.

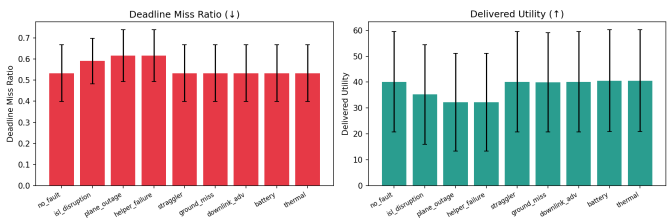

Fig. 2. ORDI robustness profile across eight fault types.

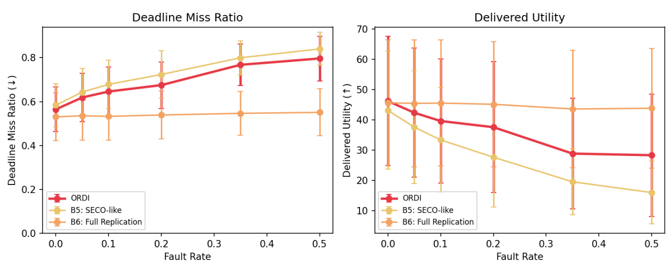

Fig. 3. Deadline miss ratio vs. fault intensity. ORDI degrades gracefully where non-redundant scheduling collapses.

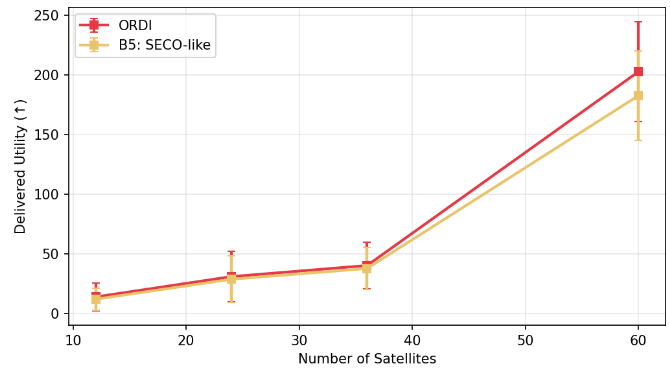

Fig. 4. Scalability with constellation size.

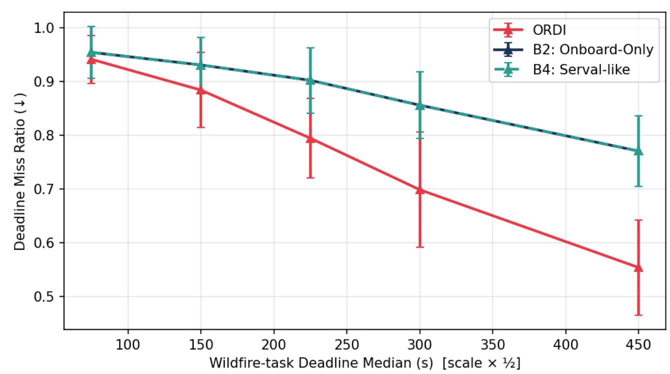

Fig. 5. Deadline tightness sweep.

every point. It misses 94.2% versus 95.5% at 150 s and 55.4% versus 77.1% at 900 s; its utility advantage grows from 17% to 60%. Mean replicas rise from 0.09 to 0.48 as longer deadlines make distributed placements and backups feasible.

G. Pricing Redundancy with λ_R

Sweeping λ_R from 0 to 2 (Fig. 6) moves mean replicas from 0.48 to 0.30 while the planned miss ratio remains 69.9%. Delivered utility falls from 28.4 to 26.8 and energy from 144 to 91 J, showing that λ_R provides a direct confidence–energy trade-off. We use λ_R=0.05 throughout, near the transition before the curve saturates at 0.1.

H. Backup-Cap Ablation

To isolate whether the one-backup cap leaves useful reliability on the table, we set λ_R=0, permit up to four total replicas, and sweep $b_{\max}$ from 0 to 3 (Fig. 7). The first backup cuts realized misses from 73.9% to 72.3% and raises utility from 26.8 to 28.4. A second reaches 72.2% misses and 28.44 utility; a third adds no measurable reliability and only 0.003 utility. Energy grows from 91 J with no backup to 144, 169, and 179 J. Thus one backup captures 93% of the maximum observed miss reduction, supporting $b_{\max}$=1 as the default.

I. Correlated Failures

Independence (Eq. 2) is the standard analytical convenience that orbital geometry breaks. We disable one or two entire orbital planes mid-horizon (Fig. 8). With plane-disjoint backups, ORDI realizes 59.8% misses for one plane and 61.3% for two, versus 56.3% and 57.7% for

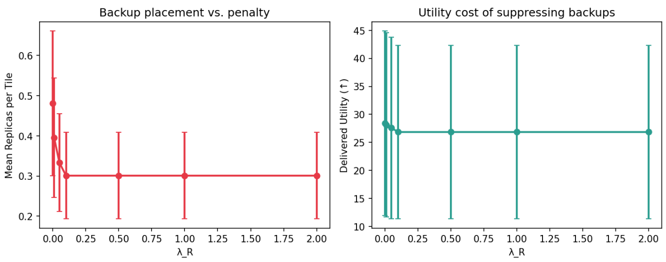

Fig. 6. Replication penalty sweep: λ_R is an effective redundancy dial.

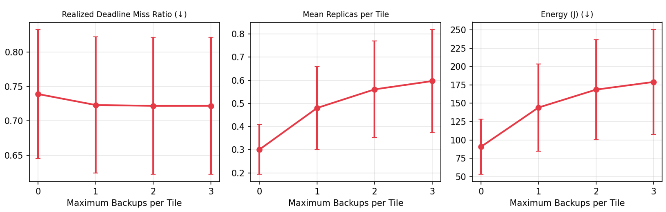

Fig. 7. Backup-cap ablation over 16 seeds with λ_R=0.

full replication. That 3.5–3.6-point gap comes with 44–45% fewer replicas and 44–45% less energy, quantifying the price of the final reliability increment.

J. Greedy vs. ILP Optimality

On small instances solvable exactly, greedy ORDI and the ILP reference produce identical planned miss ratios (88.9%). Greedy delivers slightly higher utility (4.49 vs. 4.40) while using 5.8× less ISL traffic and 32% less energy. The ILP optimizes modeled expected reliability, but its additional redundancy does not improve realized execution in these contact-constrained instances.

K. Real-World Wildfire Case Study

Scenario. We replace the synthetic input drivers with three measured datasets: CelesTrak TLEs for 24 Planet Flock/SkySat spacecraft [19], 387 in-view NASA FIRMS active-fire detections [20], and BUPT-1 battery and temperature telemetry [4]. Each detection supplies a real time, location, and fire-radiative-power-derived urgency; deadlines bracket operational wildfire-alert latency [21]. TLEs determine the contact graph, while telemetry replay replaces the RC state model. Eight seeds vary only quantities absent from these datasets: the covering source satellite, tile-level compute jitter, telemetry phase, and reliability outcomes.

Outcome. Table II preserves ORDI's utility advantage and gives it the lowest planned miss ratio among the reported methods. Full replication lowers the realized miss ratio from 47.2% to 44.9%, but uses 1.9× the replicas, 2.0× the energy, and 13.8× the ISL traffic (2.79 vs. 0.20 Gb). Direct downlink has a 48.4% planned miss ratio. Thus selective redundancy remains effective when orbital geometry, urgent request locations, and platform state all come from operational data rather than synthetic generators.

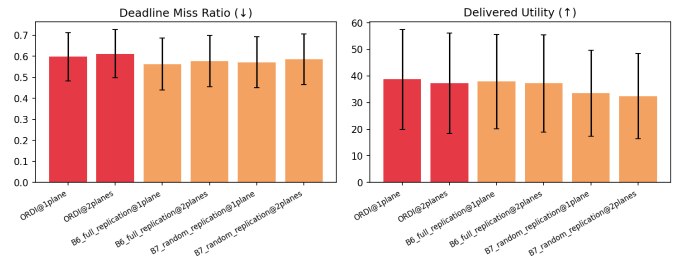

Fig. 8. Correlated plane-wide outages: plane-disjoint backups approach full-replication reliability at lower redundancy.

TABLE II
Real-world wildfire case study, mean over eight seeds.

Algorithm	Miss	Utility	Energy (J)	Repl.
B1 direct	0.484	1242.0	31	0
B5 SECO-like	0.466	614.7	254	0.53
B6 full repl.	0.426	781.3	545	1.15
ORDI	0.413	1311.0	279	0.60

VI. Discussion and Limitations

Fidelity. Our evaluation is simulation calibrated with in-orbit measurements, not a flight experiment. The compute profiles for the four workloads are benchmark-derived; we plan hardware-in-the-loop measurement on Atlas 200DK / Ascend-class accelerators, matching our calibration payload, to replace them and to validate the thermal-throttling model against device behavior. Reliability parameters. Link success probabilities are literature-typical constants; weather-dependent downlink models would sharpen the adverse cases. Coordination. ORDI assumes the source satellite has epoch-fresh constellation state; quantifying staleness tolerance, and a decentralized variant, is future work. Coding. We compare against an MDS-style baseline but do not explore coded partial redundancy (fractional parity tiles), which may dominate replication for large tile counts.

Deployment. ORDI admits two deployment models. Contact windows are deterministic given ephemerides, so a ground operator can precompute epoch graphs and uplink them with tasking; only the per-epoch assignment loop, sub-second at 60 satellites and dominated by dense array lookups after the routing sweeps, must run onboard. Alternatively, the full scheduler can run on the source satellite, since the Tiansuan logs show the assignment workload is negligible next to a single tile inference. Replanning is local by construction: it touches only tiles whose replicas failed, so a recovery decision does not require re-coordinating the constellation. Generality. Nothing in the design is specific to the Walker pattern or the two-station ground segment; the contact graph abstracts both. Denser ground segments shrink ℓ^{down} and shift the bottleneck toward compute, which strengthens the case for distribution while leaving the redundancy calculus unchanged.

VII. Conclusion

ORDI treats redundancy as a priced, orbit-aware resource rather than a blanket policy. One fault-disjoint backup captures 93% of the maximum observed miss reduction in the backup-cap ablation. Under a one-plane outage, selective placement stays within 3.5 percentage points of full replication's realized miss ratio with 44% less energy, while the greedy scheduler uses 5.8× less ISL traffic than the ILP reference. Calibration with public in-orbit COTS measurements grounds these conclusions in flight-measured power, compute, and battery behavior.

We see three immediate extensions. First, hardware-in-the-loop calibration: replacing benchmark-derived workload profiles with per-power-mode latency and energy measurements on Jetson-class flight-representative accelerators, validating the thermal-throttling model against observed frequency scaling. Second, coded partial redundancy: fractional parity tiles between one replica and two may dominate both on the energy-reliability frontier for large tile grids. Third, decentralization: bounding how stale a satellite's view of constellation state can be before selective redundancy loses its advantage, and gossiping only the state that the marginal-gain test actually consumes. Code, declarative fault specifications, and all experiment configurations are available for reproduction.

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Appendix A
Full Simulation Parameters

Table IV lists every parameter of the simulator, grouped by subsystem. Satellite hardware parameters are not assumed: they are parsed at runtime from the public Tiansuan BUPT-1 Atlas 200DK in-orbit logs [4] (inference throughput from the 4-thread image-classification log, power from telemetry, battery capacity from published charge curves).

TABLE III
Complete core-comparison results, mean over 60 seeds. Cvg. = partial coverage; Rec. = recovery latency; Util% = helper utilization; Obj. = modeled objective; Repl. = mean replicas per tile.

Algorithm	Miss	Utility	Cvg.	Rec. (s)	ISL (Mb)	Downlink (Mb)	Energy (J)	Util%	Obj.	Repl.
B1 direct	0.951	8.0	0.049	305.6	0	8.27	0.41	0	8.0	0
B2 onboard	0.853	17.2	0.147	682.6	0	11.91	45.6	0.001	17.2	0.15
B3 compress	0.853	17.6	0.147	682.6	0	3.58	4.7	0.001	17.6	0.15
B4 Serval-like	0.853	17.2	0.147	683.0	0	11.91	45.6	0.001	17.2	0.15
B5 SECO-like	0.705	22.8	0.295	721.6	60.4	25.10	94.8	0.001	22.8	0.30
B6 full repl.	0.705	24.4	0.295	833.9	102.4	25.10	171.9	0.003	20.1	0.54
B7 random repl.	0.705	23.6	0.295	878.5	124.8	25.10	175.4	0.003	19.2	0.55
B8 CoCoI-like	0.705	24.5	0.295	836.5	104.6	25.10	174.9	0.003	20.0	0.55
ORDI	0.702	26.0	0.298	700.3	41.0	25.17	103.0	0.002	25.6	0.32

TABLE IV
Complete simulation parameters.

Group	Parameter	Value
Constellation	Pattern	Walker, 6 planes × 6 sats
	Altitude / period	550 km / 5760 s
	Horizon	17,280 s (288 epochs of 60 s)
	Ground stations	Fairbanks, Greenwich
	GS min. elevation	25° (all benchmarks)
	ISL maximum range	4,000 km
	ISL rate	200 Mb/s
Tasks	Arrival process	Poisson, 6 tasks/orbit
	FOV gating	600 km radius, 100 targets
	Deadlines	log-normal, σ=0.6
	Tile size	128×128×3 uint8
Scheduler	λ_E (energy)	10^{-5} utility/J
	λ_C (comm)	10^{-12} utility/wt-bit
	λ_R (replication)	0.05 utility/replica
	α (freshness)	0.002 s^{-1}
	Replica cap $r_{\max}$	2 (primary + 1 backup)
Reliability	π_{isl}	0.97
	π_{down}	0.92 (0.70 adverse)
	π_{node}	0.98
	Straggler trigger	1.5× expected latency
Satellite (measured, Atlas 200DK BUPT-1 [4])	Compute rate	9.67 GFLOP/s
	Idle power	7.78 W
	Compute power (incr.)	2.41 W
	Comms power (incr.)	2.49 W
	Battery capacity	118.87 Wh
	Battery floor	15%
	Solar harvest (avg.)	14.43 W
	Thermal limit / max obs.	80°C / 57°C
	Thermal RC constant	300 s

Appendix B
Complete Core-Comparison Results

Table III reports all ten metrics of the core comparison, mean over 60 seeds (Table I shows the headline columns). B6–B8 lose 4–5 objective points to the replication and congestion penalties of (3), which ORDI largely avoids.

Appendix C
Fault Injection Details

Faults are declarative specifications resolved against the built constellation at runtime. The eight classes and their mechanics: ISL disruption removes a specific ISL edge from $E(t)$ for the fault duration; orbital-plane outage sets A_i=0 for every satellite in one plane; helper failure sets A_i=0 for one satellite, triggering replanning of its assigned tiles; straggler scales the helper's C_i by 0.1× during execution; ground-contact miss removes a downlink window; adverse downlink drops an aggregator's downlink success probability to the adverse value (0.70); battery shortage drains B_i below the 15% floor, forcing A_i=0 until recharge; thermal throttling raises Θ_i past the limit, reducing C_i to 25%.

The fault-intensity sweep draws, per epoch with probability equal to the fault rate, one fault uniformly from {helper failure, straggler, battery shortage, thermal throttle, ISL disruption} on a random satellite, with durations of 1–3 epochs. The correlated-failure experiment instead disables whole planes at mid-horizon, with backups constrained to plane-disjoint placement.

TABLE V
Per-tile workload profiles (four EO inference types). D = median deadline at reference scale.

Type	Model	In	Out	GFLOP	u	D (s)
Wildfire	MobileNetV2	48 kB	1 kB	0.3	1.0	300
Ship	YOLOv5-n	48 kB	5 kB	0.9	0.8	600
Cloud	U-Net (lt.)	48 kB	50 kB	1.2	0.5	1200
Change	Siamese CNN	96 kB	10 kB	1.8	0.9	480

Appendix D
Per-Experiment Configurations

Table VI gives the exact configuration of every experiment. All sweeps hold the Section V-A defaults except the swept variable; matched seeds are used across algorithms within each sweep point.

TABLE VI
Experiment configurations.

Exp.	Swept variable	Algorithms	Seeds
Core (V-B)	none	all 9	60
Fault types	8 fault classes	ORDI	16
Intensity (V-D)	rate $\in \{0, .05, .1, .2, .35, .5\}$	ORDI, B5, B6	48
Scalability	$N \in \{12, 24, 36, 60\}$	ORDI, B5	16
Deadline	scale $\in \{150..900\}$ s	ORDI, B2, B4	16
λ_R sweep	$\{0, .01, .05, .1, .5, 1, 2\}$	ORDI	16
Backup cap (V-H)	$b_{\max}\in\{0, 1, 2, 3\}$, λ_R=0	ORDI	16
Correlated (V-I)	1–2 plane outages	ORDI, B6, B7	16
ILP gap (V-J)	9/12 sats, 3/orbit	greedy, ILP	12 each
Real (V-K)	Planet TLE + FIRMS + telemetry	all 9	8